

Extracted WXT and AQT Production Dataset

Status

The combined WXT and AQT Level 0 Parquet extraction completed successfully on **2026-08-26**. All requested dates were exported and finalization completed cleanly.

Summary Contents

The dataset contains **55,038,045,135** long-format rows in Parquet format:

Sensor	VSNs	Instruments	Variables	Fact rows	Actual timestamp coverage
vaisala-wxt536	22	22	11	54,787,131,902	2023-05-05T03:30:10.267826229Z to 2025-12-16T20:14:11.123246352Z
vaisala-aqt530	15	15	12	250,913,233	2023-05-05T03:33:05.096190631Z to 2025-12-16T20:15:13.458363039Z
Total	37	37	23	55,038,045,135	—

Catalog Totals

- **23** sensor-level variables
- **383** instrument-variable combinations
- **1,756** deterministic series

Layout and Records

Facts use the following Hive layout:

```
facts/sensor=<sensor>/vsan=<VSN>/instrument=<instrument_id>/date=YYYY-MM-DD/part-*.parquet
```

Each fact preserves the nanosecond UTC timestamp, sensor, VSN, instrument ID, measurement, field, typed value, and deterministic series ID. Complete retained Influx tags are stored once in the `_series` Parquet metadata rather than being repeated in every fact row.

Extraction Performance

- **Duration & Hardware:** The initial extraction used **8 shard workers** and took **141 hours, 20 minutes, 17 seconds**.
- **Resource Usage:** Peak resident memory was approximately **2.76 GiB**.
- **Resume Log:** A timestamp conversion error occurred only during final catalog creation; no fact data was lost.

Variable Inventory

These ranges are the earliest and latest timestamps represented by each variable across the extracted dataset.

AQT Variables

Measurement	Field	Description	Units	VSNs	Rows	Start UTC	End UTC
<code>aqt.env.humidity</code>	value	Ambient Relative Humidity	percent relative humidity	15	20,909,442	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
<code>aqt.env.pressure</code>	value	Ambient Atmospheric Pressure	hPa	15	20,909,441	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
<code>aqt.env.temp</code>	value	Ambient Temperature	degrees Celsius	15	20,909,447	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
<code>aqt.gas.co</code>	value	Carbon Monoxide Gas Concentration	ppm	15	20,909,435	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
<code>aqt.gas.no</code>	value	Nitric Oxide Gas Concentration	ppm	15	20,909,436	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
<code>aqt.gas.no2</code>	value	Nitrogen Dioxide Gas Concentration	ppm	15	20,909,439	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z

Measurement	Field	Description	Units	VSNs	Rows	Start UTC	End UTC
aqt.gas.ozone	value	Ozone Gas Concentration	ppm	15	20,909,433	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
aqt.house.datetime	value	UTC time in YYYY-MM-DDTHH:MM:SS format	UTC time	15	20,909,429	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
aqt.house.uptime	value	Time in seconds since instrument startup	seconds	15	20,909,428	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
aqt.particle.pm1	value	Particulate Matter < 1 µm	µg/m ³	15	20,909,436	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
aqt.particle.pm10	value	Particulate Matter < 10 µm	µg/m ³	15	20,909,433	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z
aqt.particle.pm2.5	value	Particulate Matter < 2.5 µm	µg/m ³	15	20,909,434	2023-05-05T03:33:05Z	2025-12-16T20:15:13Z

WXT Variables

Measurement	Field	Description	Units	VSNs	Rows	Start UTC	End UTC
wxt.env.humidity	value	Ambient Relative Humidity	percent	22	6,009,703,719	2023-05-05T03:30:10Z	2025-12-16T20:14:11Z
wxt.env.pressure	value	Ambient Atmospheric Pressure	hectoPascal	22	6,009,703,550	2023-05-05T03:30:10Z	2025-12-16T20:14:11Z
wxt.env.temp	value	Ambient Temperature	degree Celsius	22	6,009,704,036	2023-05-05T03:30:10Z	2025-12-16T20:14:11Z
wxt.hail.accumulation	value	Hail Accumulation	hits/cm ²	10	1,562,254,433	2023-05-07T02:21:21Z	2025-12-16T20:01:57Z
wxt.heater.status	value	Heater Status	unitless	18	4,707,271,864	2024-10-22T20:21:27Z	2025-12-16T20:14:11Z

Measurement	Field	Description	Units	VSNs	Rows	Start UTC	End UTC
wxt.heater.temp	value	Heater Temperature	degree Celsius	21	6,009,376,932	2023-05-05T03:30:10Z	2025-12-16T20:14:11Z
wxt.heater.volt	value	Heater Voltage	volts	21	6,009,376,762	2023-05-05T03:30:10Z	2025-12-16T20:14:11Z
wxt.rain.accumulation	value	Rain Accumulation	millimeters	22	6,010,976,192	2023-05-05T03:30:10Z	2025-12-16T20:14:11Z
wxt.voltage.supply	value	Supply Voltage	volts	1	433,027,607	2024-06-20T22:49:31Z	2025-12-16T20:01:57Z
wxt.wind.direction	value	Wind Direction	degrees	22	6,012,868,573	2023-05-05T03:30:10Z	2025-12-16T20:14:11Z
wxt.wind.speed	value	Wind Speed	meters per second	22	6,012,868,234	2023-05-05T03:30:10Z	2025-12-16T20:14:11Z

VSN and Instrument Coverage

Coverage is the interval between the first and last extracted timestamp; it does not imply that every point inside the interval is present.

AQT VSN Coverage

VSN	Instrument ID	Vars	Series	Rows	Start UTC	End UTC	Coverage Days
W039	W039--vaisala-aqt530--core--5b8d3ce9a8bc	12	24	6,108	2023-05-05T03:33:05Z	2023-05-05T12:15:04Z	0.362
W069	W069--vaisala-aqt530--core--7d3281693666	12	24	13,073,943	2025-03-08T00:22:23Z	2025-12-12T19:45:13Z	279.808
W08B	W08B--vaisala-aqt530--core--32060d89ddde	12	36	19,961,471	2024-09-25T14:33:51Z	2025-12-12T19:08:48Z	443.191

VSN	Instrument ID	Vars	Series	Rows	Start UTC	End UTC	Coverage Days
W08D	W08D--vaisala-aqt530--core--df6b0090a23b	12	96	37,725,560	2023-05-05T15:40:15Z	2025-12-16T20:14:30Z	956.190
W08E	W08E--vaisala-aqt530--core--0f892acf70d0	12	108	32,435,393	2023-12-13T19:00:22Z	2025-12-16T20:14:54Z	734.052
W095	W095--vaisala-aqt530--core--e163084cea0c	12	24	13,220,039	2025-03-14T17:10:10Z	2025-12-16T20:15:00Z	277.128
W096	W096--vaisala-aqt530--core--3f57d37437e3	12	48	26,357,539	2024-06-21T01:50:32Z	2025-12-16T20:15:13Z	543.767
W098	W098--vaisala-aqt530--core--b0f19aecf20a	12	24	1,864,134	2025-04-18T01:30:26Z	2025-06-06T04:44:12Z	49.135
W099	W099--vaisala-aqt530--core--94b6cd87b5de	12	60	37,486,629	2023-12-05T22:11:05Z	2025-12-16T20:15:13Z	741.920
W09D	W09D--vaisala-aqt530--core--af3057623701	12	36	14,646,248	2025-02-25T18:14:28Z	2025-12-12T19:38:20Z	290.058
W09E	W09E--vaisala-aqt530--core--5dc161afeff8	12	36	13,416,623	2024-07-23T00:32:34Z	2025-12-12T19:34:58Z	507.793
W09F	W09F--vaisala-aqt530--core--b4389df0426d	12	12	7,716	2024-07-26T14:17:14Z	2025-02-27T23:56:09Z	216.402
W0A0	W0A0--vaisala-aqt530--core--895f71263bd4	12	24	17,947,962	2024-11-19T23:20:20Z	2025-12-16T16:43:25Z	391.724
W0A1	W0A1--vaisala-aqt530--core--7655b8cfb9bf	12	24	14,152,604	2025-02-21T00:20:15Z	2025-12-16T20:14:39Z	298.829
W0A4	W0A4--vaisala-aqt530--core--15eae7f9137	12	60	8,611,264	2024-06-27T20:40:28Z	2025-12-16T20:14:04Z	536.982

WXT VSN Coverage

VSN	Instrument ID	Vars	Series	Rows	Start UTC	End UTC	Coverage Days
W01B	W01B--vaisala-wxt536--core--1b06dce7ed4d	9	36	2,455,527,068	2023-10-19T00:37:10Z	2025-12-12T18:41:17Z	785.753
W039	W039--vaisala-wxt536--core--8b10123cd91c	8	8	3,021,256	2023-05-05T03:30:10Z	2023-05-05T12:15:09Z	0.365
W057	W057--vaisala-wxt536--core--81ca8e85f911	9	46	79,241,027	2023-05-06T19:20:19Z	2023-08-31T13:05:27Z	116.740
W067	W067--vaisala-wxt536--core--96608f5c339f	10	67	354,754,543	2023-10-19T01:01:30Z	2025-12-16T20:10:49Z	789.798
W069	W069--vaisala-wxt536--core--862be7fe459e	7	24	7,316,497	2024-10-07T22:21:46Z	2025-12-01T01:33:27Z	419.133
W06A	W06A--vaisala-wxt536--core--f6c1fd64a63b	9	36	1,504,243,062	2024-10-21T14:00:08Z	2025-10-24T19:02:59Z	368.210
W06C	W06C--vaisala-wxt536--core--4fcd0ec203d1	9	60	2,625,646,591	2024-11-20T03:50:09Z	2025-12-16T20:14:11Z	391.683
W071	W071--vaisala-wxt536--core--463d13083d94	9	18	1,836,946,600	2025-02-24T21:30:08Z	2025-10-07T22:39:03Z	225.048
W08B	W08B--vaisala-wxt536--core--42f10dc622f2	9	74	3,444,183,324	2024-09-25T14:31:52Z	2025-12-12T19:08:58Z	443.192
W08D	W08D--vaisala-wxt536--core--934c67f6166a	9	154	6,150,739,142	2023-05-05T16:20:07Z	2025-12-16T20:00:50Z	956.153
W08E	W08E--vaisala-wxt536--core--e79753d1a123	9	131	4,117,030,359	2023-05-19T17:10:08Z	2025-12-16T20:01:31Z	942.119
W095	W095--vaisala-wxt536--core--792d997c77e2	10	33	2,627,296,648	2025-03-14T17:10:11Z	2025-12-16T20:01:21Z	277.119
W096	W096--vaisala-wxt536--core--cbc808667e24	11	102	4,677,125,543	2024-06-20T22:49:31Z	2025-12-16T20:01:57Z	543.884
W097	W097--vaisala-wxt536--core--649ca0b6f6e4	9	20	2,414,315,755	2025-01-	2025-12-	318.975

VSN	Instrument ID	Vars	Series	Rows	Start UTC	End UTC	Coverage Days
					31T20:50:12Z	16T20:13:35Z	
W098	W098--vaisala-wxt536--core--aa22a8ace90f	10	32	238,050,683	2025-04-17T22:57:26Z	2025-06-06T04:44:30Z	49.241
W099	W099--vaisala-wxt536--core--6c922d091b8d	9	70	6,367,106,700	2023-12-05T22:10:10Z	2025-12-16T20:02:05Z	741.911
W09D	W09D--vaisala-wxt536--core--23a9906b4c45	10	32	2,940,841,702	2025-02-25T18:12:28Z	2025-12-12T19:38:36Z	290.060
W09E	W09E--vaisala-wxt536--core--d09cea632c7b	10	31	2,633,312,391	2024-07-26T21:28:27Z	2025-12-12T19:35:11Z	503.921
W09F	W09F--vaisala-wxt536--core--15bba9089188	9	9	109,016	2024-07-26T14:07:59Z	2024-07-26T17:48:16Z	0.153
W0A0	W0A0--vaisala-wxt536--core--3981956aa520	9	49	3,357,907,277	2024-11-19T23:20:13Z	2025-12-16T16:43:27Z	391.724
W0A1	W0A1--vaisala-wxt536--core--ee3274a084ad	10	33	2,838,193,809	2025-02-21T00:20:06Z	2025-12-16T20:01:06Z	298.820
W0A4	W0A4--vaisala-wxt536--core--354abfb39272	9	55	4,114,222,909	2024-06-27T20:30:06Z	2025-12-16T20:01:00Z	536.980